Week 1: Programming Fundamentals

Session 1: Basics & Conditionals

- Data types, variables, input/output
- Operators and expressions
- if-else, nested if-else, switch-case
- Flowcharts and logic building

Session 2: Loops & Pattern Printing

- For, while, do-while loops
- Nested loops
- Pattern-based problems
- Loop-based complexity

Session 3: Functions & Method Concepts

- Function declaration, parameters, return types
- Call by value/reference
- Method scope and recursion intro

Session 4: Time Complexity & Flowcharts

- Big O Notation
- Time complexity of common code patterns
- Analyzing loop + nested structures
- Intro to dry run strategies

Session 5: Introduction to Recursion

- Recursive calls & call stack
- Base and recursive case
- Simple problems: factorial, power
- Recursion tracing

Week 2: Arrays, Strings & Sorting

Session 6: 1D Arrays – Operations & Applications

- Declaration and initialization
- Traversal, insertion, deletion
- Max/min, prefix sum, reversal

Session 7: 2D Arrays – Traversal & Matrix Operations

- Row/column/diagonal traversal
- Transpose and rotation
- Spiral matrix and boundary traversal

Session 8: Time and Space Complexity Analysis

- Best/worst/average case
- Space vs time trade-offs
- Practical analysis on loops and recursion

Session 9: String Manipulation & Problem Solving

- String creation and operations
- Palindrome and anagram check

- String reversal, substring extraction

Session 10: Sorting – Bubble, Selection & Insertion

- Dry run examples
- Time complexities
- When and why to use each

Week 3: Binary Search & Recursion

Session 11: Fundamentals of Binary Search

- Mid-point logic, base condition
- Binary Search on sorted arrays

Session 12: Applications of Binary Search

- First/last occurrence
- Count of occurrences
- Search in rotated array

Session 13: Recursive Patterns – Fibonacci, Power, Factorial

- Top-down recursion
- Stack depth
- Multiple return calls

Session 14: Recursion with Subsets and Permutations

- Generating subsets

- All permutations
- Bitmasking intro

Session 15: Backtracking Intro

- Decision trees
- Pruning invalid paths
- N-Queens problem

Week 4: Sorting, OOPs & Linked Lists

Session 16: Merge Sort – Divide and Conquer

- Splitting arrays
- Merge step
- Merge sort with recursion

Session 17: Quick Sort – Partition and Optimization

- Lomuto/Hoare partition
- Worst vs best case
- In-place sorting

Session 18: OOPs Concepts – Classes & Inheritance

- Class/object basics
- Inheritance types
- Access modifiers

Session 19: OOPs Concepts – Abstraction & Interfaces

- Abstraction vs encapsulation
- Abstract classes and interfaces
- Constructor overloading

Session 20: Linked List – Basics & Implementations

- Singly linked list
- Insert/delete operations
- Traversal techniques

Week 5: Stacks, Queues & Hashing

Session 21: Stack Data Structure – Operations & Use Cases

- Push/pop/peek
- Stack using arrays
- Expression validation

Session 22: Stack Interview Problems

- Next greater element
- Stock span
- Valid parentheses

Session 23: Queue Variants & Implementations

- Queue operations
- Circular queue
- Queue using stacks

Session 24: Queue-Based Interview Problems

- First non-repeating character
- Sliding window maximum
- Rotten oranges (BFS)

Session 25: HashMaps & HashSets in Depth

- Hash function and collision
- Frequency maps
- Two-sum, union/intersection

Week 6: Trees & Binary Search Trees

Session 26: Binary Trees – Structure & Traversals

- Preorder/inorder/postorder
- Recursive/iterative traversal
- Height/depth

Session 27: Tree Views – Level, Vertical, Zigzag

- Level order (BFS)
- Left/right/top/bottom view
- Diagonal and zigzag traversal

Session 28: Binary Search Trees – Insert, Delete, Search

- Insert/search/delete logic
- Validate BST

- Min/max in BST

Session 29: Tree Problems – LCA, Diameter, Mirror Tree

- Lowest Common Ancestor
- Diameter and height
- Mirror and symmetric trees

Session 30: Tree Practice – Recursive Techniques

- Practice problems
- Space optimization with Morris traversal
- Tree-to-DLL conversion

Week 7: Heaps, Prefix Sum, Sliding Window, Primes

Session 31: Heaps – Min/Max & Priority Queues

- Heapify, insert/delete
- Heap sort
- Priority queue applications

Session 32: Prefix Sum Techniques

- Prefix sums in arrays
- Difference arrays
- Range sum queries

Session 33: Sliding Window – Fixed & Dynamic Windows

- Max sum subarray
- Longest substring with K distinct chars
- Sliding window in strings

Session 34: Prime Numbers – Efficient Computation

- √N primality test
- Prime factorization
- Prime count up to N

Session 35: Sieve of Eratosthenes & Number Theory

- Classic and segmented sieve
- Prime ranges
- Modulo arithmetic basics

Week 8: Binary Search II, Backtracking & Greedy

Session 36: Advanced Binary Search Techniques

- Search in infinite array
- Binary search on answer problems
- Lower_bound and upper_bound

Session 37: Backtracking Problems – N-Queens, Maze, Sudoku

- State space tree
- Constraint-based pruning
- Sudoku solver

Session 38: Greedy Algorithms – Activity Selection & Knapsack

- Activity selection
- Fractional knapsack
- Sorting-based decisions

Session 39: Greedy Algorithms – Job Scheduling & Gas Station

- Job sequencing
- Gas refill problem
- Sorting + greedy merges

Session 40: Greedy Applications – Minimum Platforms, Intervals

- Overlapping intervals
- Huffman encoding
- Interval covering problems

Week 9: Dynamic Programming

Session 41: DP Foundations – Memoization & Tabulation

- Top-down vs bottom-up
- State definition and recurrence
- Fibonacci, climbing stairs

Session 42: Classic Problems – Knapsack, Subset Sum

- 0/1 Knapsack
- Subset sum

- Count subsets with given sum

Session 43: Coin Change & Minimum Ways

- Coin combinations
- Minimum coins
- Unbounded knapsack

Session 44: LIS, LCS & Matrix-Based DP

- Longest Increasing Subsequence
- Longest Common Subsequence
- DP in grids/matrices

Session 45: DP Patterns for Interview Success

- Choice diagrams
- Space optimization
- Practice mix problems

Week 10: Graphs & Tries

Session 46: Graph Theory – Representation & Traversal

- Adjacency list/matrix
- BFS and DFS
- Graph input patterns

Session 47: Graph Problems – Cycle Detection & Components

- Detect cycle in undirected/directed
- Connected components
- DFS forest

Session 48: Shortest Paths & Topological Sort

- Dijkstra’s algorithm
- Topological sort (Kahn's + DFS)
- Shortest path in DAG

Session 49: Tries – Insert, Search, Delete

- Trie implementation
- Insert/search/delete operations
- Use in dictionary apps

Session 50: Tries Practice – Autocomplete & Word Dictionary

- Autocomplete system
- Longest common prefix
- Word break and wildcard matching